VIDIT SHAH

Mumbai, India | Mobile: **+91 8355941192** | Email: **f20230434@pilani.bits.pilani.ac.in** | **LinkedIn**

EDUCATION

Birla Institute of Technology and Science Pilani, Pilani Campus, Pilani, Rajasthan **2023 - Present**
Bachelor of Engineering in Mechanical Engineering | **CGPA:** 8.79 | Currently 3rd Year/ 5th Semester
Prakash College of Commerce and Science, Mumbai, Maharashtra | HSC (PCM), French **2021 - 2023**

INTERNSHIPS & RELEVANT EXPERIENCE

Undergraduate Researcher, BITS Pilani, Pilani, Rajasthan, India **Jan 2025 - Present**
*Working as a **student researcher** under the guidance of Dr. Radha Raman Mishra (Assistant Professor of Mechanical Engineering).*

- Conceptualized and constructed a Proof-of-Concept setup for **Rapid Volumetric Polymerization of Optically Opaque Resin using Microwave Energy**. The process describes volumetric curing of resin using microwaves to selectively cure a rotating vat of opaque, microwave sensitive resin.
- I used **FDM & SLA Additive Manufacturing Processes** to fabricate the parts, and integrated them with electronics, using an **Arduino Uno, motor drivers and a Nema 17**. Also took apart an old microwave oven to harness the microwaves, incorporating it with a wave guide to efficiently route the microwaves to the required zone. I created an enclosure for the setup using conventional manufacturing processes such as machining and welding of sheet metal.
- Conducted systematic experiments using varying speeds to rotate the resin vat, and varying levels of power for the microwaves. The resulting data will be documented and **filed for patent** later this year.

Undergraduate Teaching Assistant, Department of Mechanical Engineering, Pilani, Rajasthan, India **Jan 2026 - May 2026**
*Working as a **Teaching Assistant** for the Course ME F221 Mechanisms & Machines.*

- Took multiple workshops teaching **MATLAB & Simulink Simscape** to 150+ 2nd year students, teaching them dynamic & static force analysis of a 4-bar linkage & its effects on the joint reaction forces. Also conducted invigilation for class tests.

Summer Intern, L&T Heavy Engineering, Larsen & Toubro Group, Surat, Gujarat, India **May 2025 - July 2025**
L&T Heavy Engineering is a leader in heavy manufacturing, supplying engineered to order equipment for various heavy industries.

- Contributed to the SMART Manufacturing group at L&T Heavy engineering, with the goal of developing novel industrial fabrication techniques to improve L&T's fabrication capabilities. The SMART Manufacturing Research group specifically deals with new age technologies such as drone tech, robotics and digital twin. My specific role involved coming up with solutions for automating Gas Tungsten Arc Welding (GTAW) Buttering operations, to improve weld quality and efficiency.
- Designed an industrial-scale 6-Axis Robotic Arm from scratch, with the intended purpose of automating GTAW Buttering operations. The arm was. The Process involved end-to-end component selection, vendor outreach, sourcing of gearboxes and motors, CAD Design in SolidWorks, and Finite Element Analysis. Also developed an initial, scaled down Proof-of-Concept with FDM 3D printing.
- Gained comprehensive industrial exposure, getting to experience various cutting edge manufacturing technologies firsthand, via frequent visits to other L&T group departments & companies, such as Casting, Forging, Energy (Turbines), Defense (Armored Vehicles Complex) and Nuclear Design.

Summer Intern, Ernst & Young (EY), Mumbai, Maharashtra, India **June 2024 - Aug 2024**
EY is a global professional services organization that provides assurance, consulting, tax, strategy, and transactions services.

- Worked on a technical consulting project for JSW Steel with the goal of optimizing their production supply chain, consisting of production forecasting and resource allocation. I collaborated with EY's data science team to understand existing operational challenges and gather data requirements for a regression-based solution.
- Worked alongside clients to develop a multivariate regression model in Python with libraries such as Pandas & NumPy. The model was able to predict sales, costs and production figures using historical company data.

POSITIONS of RESPONSIBILITY

Vice-Captain & Mechanical Engineering Lead, CRISS Robotics, Pilani, Rajasthan, India **Feb 2025 - Present**
CRISS Robotics is BITS Pilani's premier student technical & research team, working on development of a Mars Rover Prototype.

- As Vice-Captain, I was involved with **project management** & **resource allocation**. Alongside the captain, I co-lead a team of over **70 multidisciplinary engineering students**. Directed fundraising initiatives by engaging sponsors & college alumni to raise over **20,000$.**
- As the Mechanical Engineering Lead, **I directly led a team of over 25 mechanical engineering students** for the complete developmental lifecycle of our Mars Rover prototype. Within a one-year timeframe, we were able to entirely revamp our rover, with an entirely **new mechanical & scientific analysis system, A new 5-DOF Robotic Arm and a suspension system**. Emphasizing on **simulations & generative design** and using the latter to design our TPU tyres. Focused heavily on R&D efforts, coming up with **novel systems** such as **mechanical rotary encoding**. Also shifted the team towards self dependance, reducing reliance on external vendors.
- CRISS is also the **first, and only Indian team** to be invited to the **Australian Rover Challenge**, hosted by the University of Adelaide. We also take part in several annual competitions such as European Rover Challenge, International Rover Challenge, Robofest Gujarat.

VOLUNTEERING/SOCIAL WORK

Executive Committee Member, National Service Scheme, BITS Pilani **Aug 2023 - Present**

NSS is a social service initiative by the Government of India, to inculcate social responsibility in the Indian youth.

- Led Umang Collection Drive, engaging 4000+ BITS students to raise over a total of ₹30L (35,000$) over 3 years, directly funding scholarships and educational support for underprivileged children. The annual three-day donation drive consisted of unique outreach and marketing challenges, requiring problem solving and people skills.
- Provided 500+ Scholarships through Project Umang, enabling children in the rural towns around BITS Pilani with access to necessary education. Team members would carry out home visits to interact with the children & their families, to better understand the qualms of the local populous. As a part of NSS, I also had the opportunity to guide children towards their future aspirations, by means of workshops and informational seminars, where we would disseminate information on various fields, their job opportunities and the pathway to achieving further education.

RELEVANT COURSEWORK

Relevant Courses (Completed/To be completed by May 2026): Mechanics of Solids, Mechanisms & Machines, Design of Machine Elements, Advanced Mechanics of Solids, Additive Manufacturing, Digital Image Processing, Robotics, Control Systems, Vibrations & Control, CAD, Manufacturing Management, Engineering Optimization, Engines, Motors & Mobility, Heat Transfer, Prime Movers & Fluid Machines

PROJECTS

Mars Rover Prototype (SolidWorks, Fusion360, Ansys, FDM 3D Printing, Orca Slicer, CNC Milling, EDM Sheet Metal)

- Designed & Developed a Mars Rover Prototype. Personally designed assemblies such as 4-Wheel Independent Steering with worm-spur gearboxes, Rocker Suspension with custom differential bar, Hub Motor assembly, with custom TPU-95 tyres.
- Overhauled our chassis with a 2020 extrusion chassis, focusing on modularity & light weight, while maintaining rigidity.

5-Axis Robotic Arm (SolidWorks, Fusion360, Ansys, FDM 3D Printing, Orca Slicer, MATLAB, Multibody Simscape)

- Developed a 5-axis Robotic Arm. Work includes static & dynamic simulations & kinematic design. Developed custom 3D Printed Pinwheel Cycloidal drives with absolute encoders as well as a differential bevel gear wrist & leadscrew-based end-effector. The arm was designed to be over a meter long with a nominal payload of 10 Kgs.
- Developing hot-swappable end effectors, using dovetails & ball detents. The arm swaps between a 2-claw gripper or a scoop.

Soft Robotic End Effector (SolidWorks, FDM & SLA 3D Printing, Orca Slicer, Arduino IDE)

- Working on a pneumatically actuated 4-Jaw Soft Robotic End Effector, to interact with fragile & irregular objects. The End Effector comprises of 4 silicon “fingers”, capable of flexing when pressurized, to induce motion & grip. An air pump and a solenoid valve.
- The silicon fingers were developed using a molding process. LSR was poured into custom FDM Printed molds to create the required geometry. The 3-part molds also featured over molded PLA parts, inserted into silicone to provide rigidity at the input. A thin PETG slice is also inserted into the gripping face. Printed with special textures, the PETG slice is over molded to provide repeatable flexure & support.

Cable-Driven Continuum Robot (SolidWorks, Ansys, FDM 3D Printing, Orca Slicer, Arduino IDE)

- Leading Development of a tendon-driven Continuum Robot for navigating complex obstructions, with theoretically infinite DOFs, inspired by biological manipulators. Used compliant FDM printed, cable-driven segments for actuation, driven by Nema17s.
- Current development is focused on miniaturizing the design and mounting of a camera to provide visual feedback. The Robot also features provisions to mount an end effector and additionally, the arm itself can be mounted to a robotic arm, increasing it’s capabilities.

Thermal Management & Cooling Optimization of High-Energy Density EV Batteries (MATLAB Simulink, Simscape)

- Ran a case study on the battery pack design of the Jaguar I-Pace. The developed model would calculate & simulate all types of vehicle load (urban, highway, mixed), charging & recharging cycles, also featuring battery age simulations.
- Worked on high fidelity Electrochemical-Thermal battery modelling using the Battery toolbox and the Fluids Toolbox on MATLAB Simulink, to simulate a liquid cooling system. Case study developed as a part of course ME F317 Engines, Motors & Mobility.

TECHNICAL SKILLS

CAE & CAD: SolidWorks, Fusion 360, Siemens Solid Edge, MATLAB, Ansys, Multibody Simscape	**Misc:** 3D Printing (FDM & SLA), Machining, Image processing, Adobe Suite, Arduino IDE

COMPETITIONS/AWARDS

- **2nd Runner Up, International Rover Design Challenge**, International Space Robotics (SPROS) Week, 2025
- **2nd Runner Up, European Rover Challenge (Remote),** European Space Foundation, 2025
- **National Finalist, World Robot Olympiad,** for 3 consecutive years, 2016-18
- **National Runner Up, Jr. Robocon,** 2016
- **National Finalist, Robotex India,** 2019

LANGUAGES

English: Native/Bilingual, **Hindi:** Native/Bilingual, **Gujarati:** Native/Bilingual, **French:** Elementary Proficiency